

RETAILMART V3 • SQL

Subqueries Part 1 — Quick Quiz

Answer Before Looking!

Test your subquery intuition before checking the answer.

Quiz 1: What Does This Return?

EASY

ANSWER: Names of employees earning above the company-wide average salary. The inner query returns ONE number (the average). The outer compares each employee's salary to it. Classic scalar subquery in WHERE.

Question



ANSWER

```
SELECT first_name FROM stores.employees
WHERE salary > (
    SELECT AVG(salary)
    FROM stores.employees
);
```

Quiz 2: IN-Subquery vs JOIN

MEDIUM

Do these two queries return the same rows?

ANSWER: YES — same rows. Both find customers with at least one order. Performance varies: the optimizer often rewrites them to the same plan. Choose based on readability — IN-subquery is clearer when you only need rows from one table; JOIN+DISTINCT wins when you need columns from both.

Question



ANSWER

-- Query A

```
SELECT * FROM customers.customers c
WHERE c.customer_id IN (
    SELECT cust_id FROM sales.orders);
```

-- Query B

```
SELECT DISTINCT c.* FROM customers.customers c
JOIN sales.orders o
    ON c.customer_id = o.cust_id;
```

Quiz 3: The NOT IN Trap

MEDIUM

ANSWER: On RetailMart prod_id has no NULLs, so this works — 604 never-sold products. But IF the inner list contained even ONE NULL, NOT IN would return ZERO rows (the whole comparison becomes UNKNOWN). Defensive fix: WHERE prod_id IS NOT NULL in the inner query, or use NOT EXISTS (tomorrow) — it is NULL-proof.

Question

 ANSWER

```
SELECT * FROM products.products
WHERE product_id NOT IN (
    SELECT prod_id FROM sales.order_items
);
-- What if order_items had NULL prod_ids?
```


Quiz 4: Scalar Subquery Limits

HARD

ANSWER: FAILS — dept 5 has 302 employees, and a scalar subquery must return exactly ONE row, ONE column ('more than one row returned by a subquery used as an expression'). Fix: aggregate it — (SELECT MAX(salary) FROM stores.employees WHERE dept_id = 5). Now it returns one number.

Question

✓ ANSWER

```
SELECT dept_name,  
       (SELECT salary FROM stores.employees  
        WHERE dept_id = 5)  
FROM core.dim_department;  
-- What goes wrong?
```

Quiz 5: Derived Table Alias

CRAZY

What's wrong with this query?

ANSWER: Two problems. (1) Derived table has no alias — PostgreSQL requires one. (2) The aggregate column has no alias, so 'sum' (the function name) isn't directly addressable. Fix: AS total_rev outside the subquery and AS revenue inside the SUM.

Question

 ANSWER

```
SELECT * FROM (  
  SELECT store_id, SUM(net_total)  
  FROM sales.orders  
  GROUP BY store_id  
) WHERE sum > 100000;
```